

Quantum-Inspired Orthonormal Networks for Sustainable Medical Image Restoration

Sayantan Dutta

STO, GE HealthCare, Bangalore 560066, Karnataka, India.

1 Introduction

Medical image restoration is a critical problem across MRI, CT, PET, and ultrasound, where acquisition noise, low-dose protocols, hardware limitations, and modality-specific artifacts can obscure subtle anatomy and degrade downstream diagnosis or quantitative analysis. Modern CNN and transformer-based restoration models achieve strong performance [3], but often require large parameter counts, high memory, long inference time, and substantial energy consumption. These costs hinder scanner-integrated reconstruction, real-time imaging, edge deployment, and resource-constrained clinical use.

We explore a complementary direction: rather than scaling model size, we impose physically motivated structure on feature representations. By combining unitary channel mixing and orthonormal feature representations, QICNN yields energy-stable, non-redundant, information-preserving features. We validate it on clinical brain MRI denoising and preliminary low-dose CT examples, suggesting a compact modality-agnostic restoration primitive.

2 Method

QICNN adopts a compact UNet-style encoder-decoder with two quantum-inspired modules. First, *Quantum Gate* modules implement orthonormal 1×1 channel transformations, enforcing unitary, norm-preserving feature mixing. This preserves feature energy and reduces uncontrolled activation amplification or attenuation. Second, *Orthonormal Basis* layers decorrelate feature maps, encouraging non-redundant channel representations and efficient capacity usage.

This structured feature space provides a principled alternative to increasing depth, width, or attention complexity. QICNN was evaluated on clinically acquired noisy brain MRI and compared with UNet, RED-CNN [2], and HAIR [1]. Metrics included qualitative quality, GLCM texture descriptors, CNR, parameters, latency, energy, and CO₂ emission. Phantom and CatSim low-dose CT experiments assessed fidelity and preliminary cross-modality feasibility.

3 Results

Fig. 1 summarizes MRI quality, texture analysis, and efficiency. Across 20 brain MRI studies, QICNN achieved the highest GLCM energy (0.152 ± 0.056), homogeneity (0.613 ± 0.092), and CNR (6.283 ± 2.019), and the lowest dissimilarity (4.355 ± 0.673), indicating improved texture consistency and low-contrast preservation. Compared with HAIR, QICNN reduced parameters by 93.3% (1.91M vs. 28.56M), latency by 98.6% (1.14ms vs. 82.15ms), energy by 97.4% (0.63J vs. 24.23J), and CO₂ emission by 97.5%.

Fig. 2 further shows low-dose CT denoising and ablation results. CT examples demonstrate suppression of noise and streak-like artifacts while preserving structural boundaries. Ablation confirms that Quantum Gate singular values remain close to unity, validating the intended norm-preserving and decorrelating

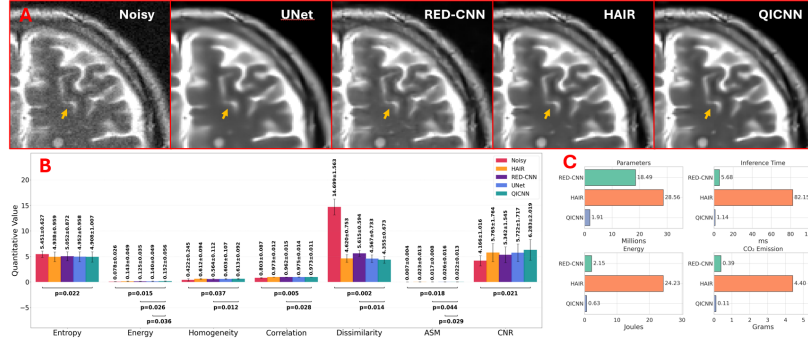


Fig. 1. QICNN performance summary. (A) MRI denoising examples show noise suppression with preserved anatomical detail. (B) GLCM/CNR analysis shows improved texture consistency and contrast. (C) Profiling shows lower parameters, latency, energy, and CO₂ emission.

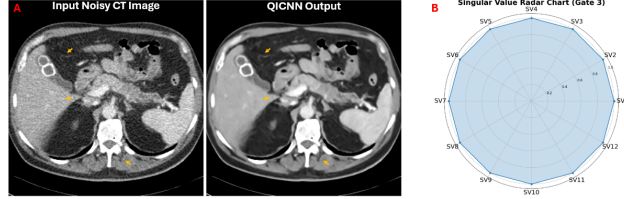


Fig. 2. Cross-modality feasibility and ablation. (A) Low-dose CT examples show noise/artifact suppression. (B) Ablation confirms unit singular values and orthonormal feature representations.

behavior. These results suggest that orthonormal feature encoding may provide a useful architectural principle beyond a single imaging modality.

4 Clinical Implications and Conclusion

Medical image restoration tools must be accurate, fast, robust, affordable, and sustainable for clinical deployment. QICNN preserves fine anatomy and improves CNR while reducing model size, latency, energy consumption, and hardware burden, supporting real-time MRI enhancement, low-dose CT denoising, scanner-integrated restoration, edge deployment, and resource-constrained imaging. More broadly, the results highlight the value of structured neural representations for medical imaging. Instead of relying only on increasingly large models, restoration networks can benefit from physically motivated constraints that preserve information, reduce redundancy, and improve energy stability. QICNN demonstrates that orthonormal and unitary constraints can provide compact, interpretable, and efficient building blocks for sustainable medical image restoration across modalities.

References

1. Cao, J., et al.: HAIR. arXiv preprint arXiv:2408.08091 (2024)
2. Chen, H., Zhang, Y., Zhang, W., Liao, P., Li, K., Zhou, J., Wang, G.: Low-dose CT via convolutional neural network. Biomedical Optics Express **8**(2), 679–694 (2017)
3. Ding, N., et al.: Source-free domain adaptation via distribution estimation. In: CVPR. pp. 7212–7222 (2022)